

File Compression Tool

Week 1 — Detailed Plan: Requirement Analysis & Project Design

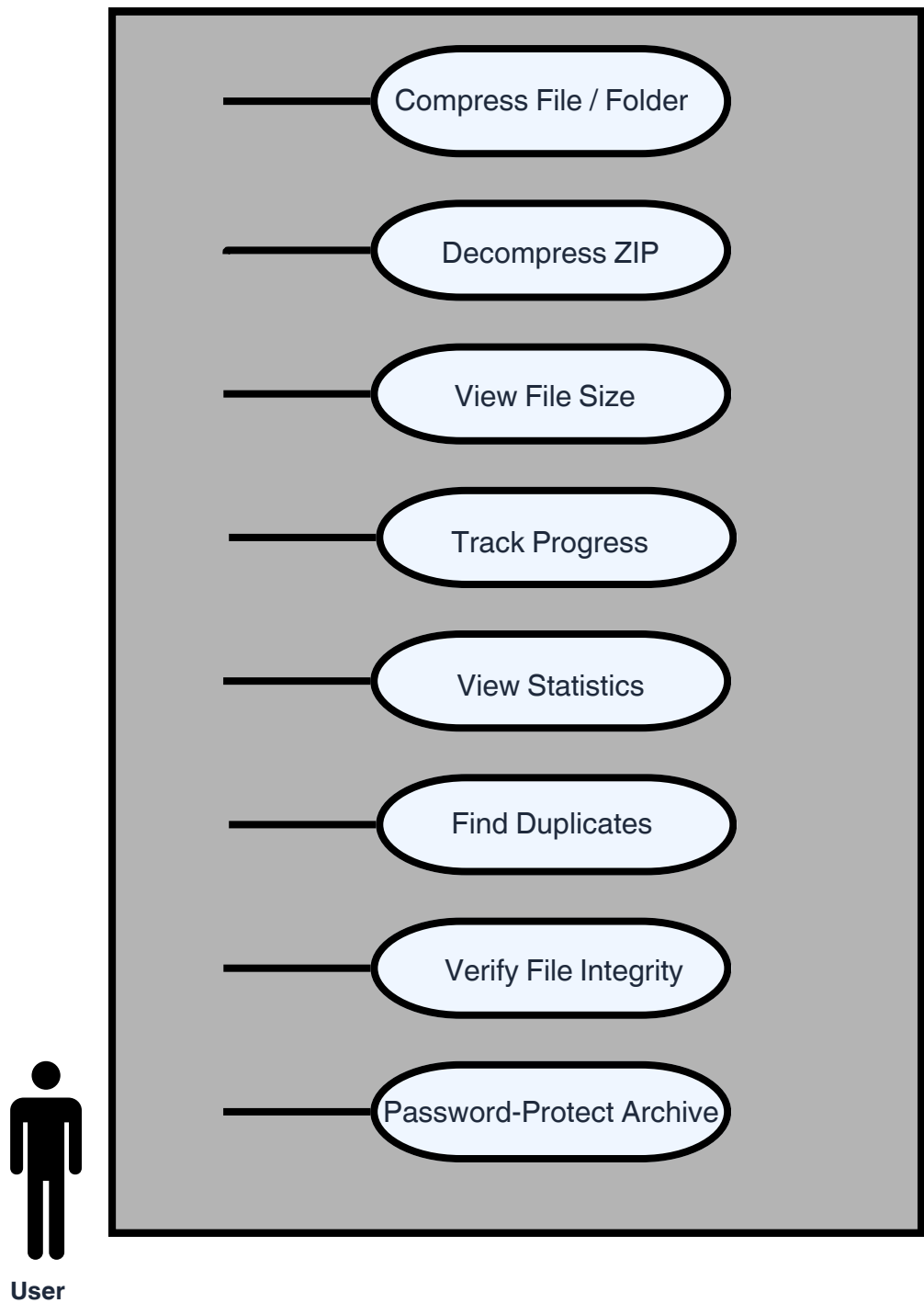
Project Information

Project: File Compression Tool
Language: Java (Swing / JavaFX)
Duration: 10 Weeks
Week: 1 — Requirement Analysis & Project Design

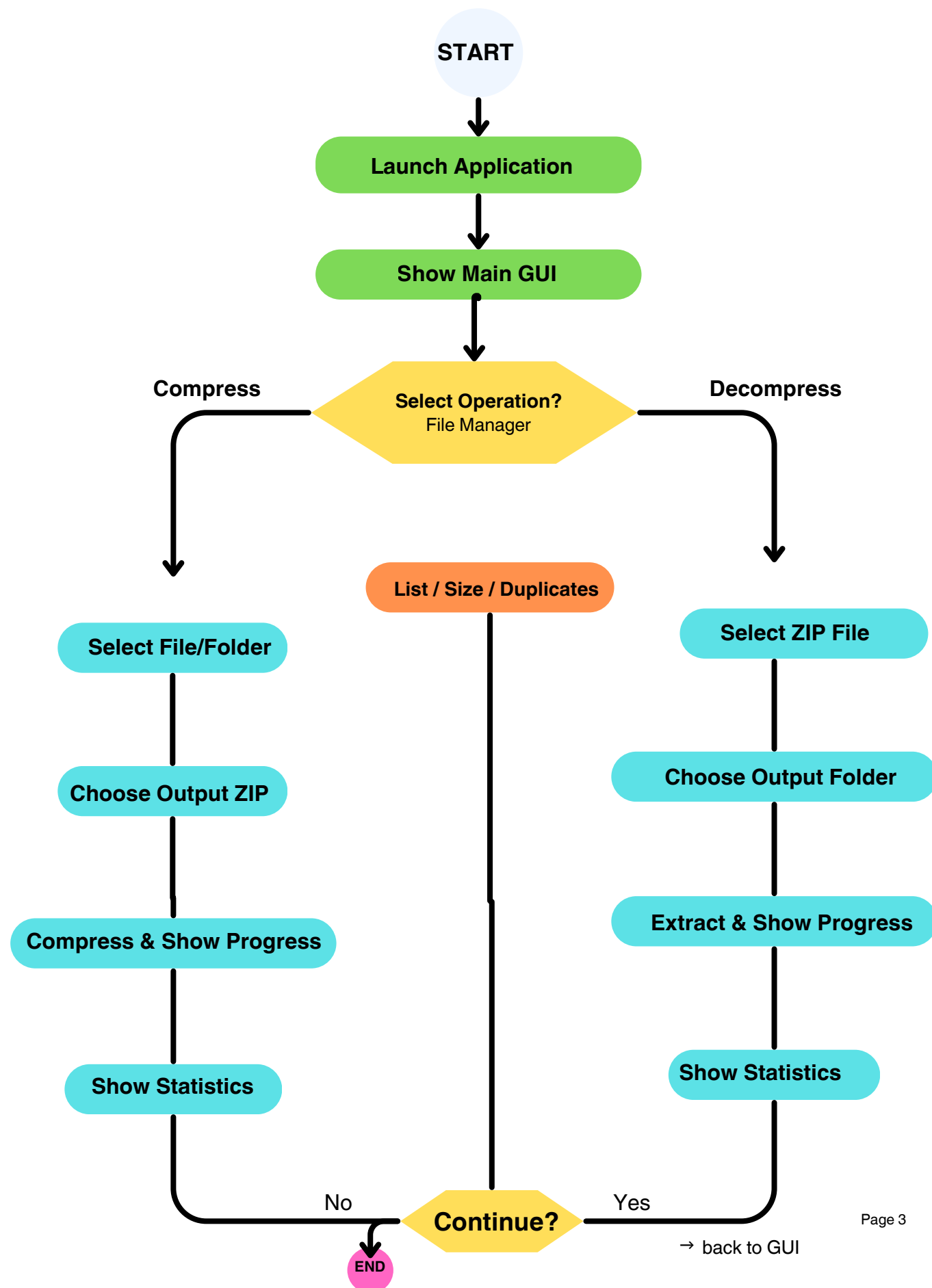
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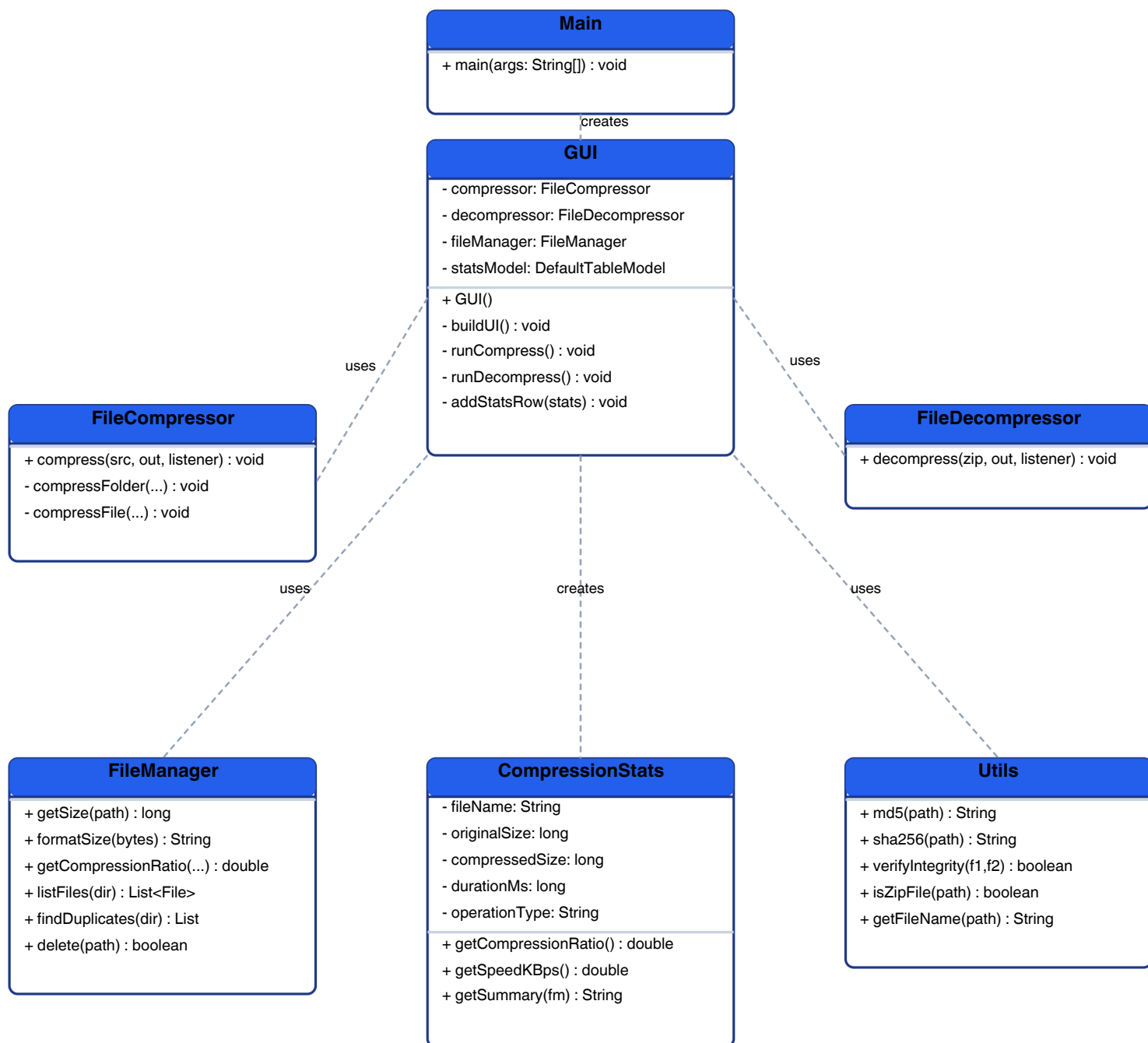
Use Case Diagram



Application Flowchart



Class Diagram



Development Environment Setup

JDK:	JavaDevelopmentKit 21
IDE:	NetBeans IDE (project base: Java Application)
Version Ctrl:	GitHub repository — commit after each session
Build:	NetBeans built-in Ant build (no Maven needed)

Planned Class Structure

```
FileCompressionTool/  
src/filecompressiontool/  
    Main.java  
    GUI.java  
    FileCompressor.java  
    FileDecompressor.java  
    FileManager.java  
    CompressionStats.java  
    Utils.java
```

Next Step — Week 2:

Begin implementing the GUI layout using Java Swing in NetBeans. Source files provided: GUI.java, Main.java, and all supporting classes.